

HENICHES'K, Ukraine

WMO: 339100

Lat: 46.1642N

Lon: 34.8072E

Elev: 15

StdP: 101.14

Time Zone: 2.00 (EUE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.3	-15.9	0.9	-12.6	-13.2	1.2	-9.8	24.7	-5.8	19.9	-2.7	4.9	0	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	31.3	22.4	30.0	22.2	28.9	21.6	24.4	29.3	23.5	28.6	22.6	27.7	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.6	27.9	21.7	16.4	27.2	20.7	15.4	26.2	73.9	29.1	70.4	28.9	66.7	27.5	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.1	13.6	-16.3	34.6	3.0	2.3	-18.4	36.3	-20.2	37.6	-21.9	38.9	-24.1	40.6		
			-16.6	26.8	3.0	1.3	-18.8	27.7	-20.5	28.4	-22.2	29.1	-24.4	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.1	0.1	4.1	10.2	16.8	21.0	24.2	23.0	17.7	11.2	5.7	0.1	(6)
(7)		DBStd	9.70	4.41	4.54	3.53	3.65	3.34	2.73	2.38	2.78	3.53	4.35	4.24	4.93	(7)
(8)		HDD10.0	1331	344	277	185	42	1	0	0	0	1	37	137	307	(8)
(9)		HDD18.3	3094	603	510	441	244	69	8	0	2	51	224	378	564	(9)
(10)		CDD10.0	1751	0	0	2	48	212	332	439	403	233	73	9	0	(10)
(11)		CDD18.3	475	0	0	0	1	21	89	181	147	33	3	0	0	(11)
(12)		CDH23.3	3562	0	0	0	3	64	552	1622	1211	106	3	0	0	(12)
(13)		CDH26.7	939	0	0	0	0	5	93	474	348	19	0	0	0	(13)
(14)	Wind	WSAvg	4.3	4.3	4.4	5.1	5.0	3.9	3.8	3.5	3.4	4.2	4.4	4.7	4.7	(14)
(15)	Precipitation	PrecAvg	391	33	31	34	32	34	42	37	32	30	26	32	38	(15)
(16)		PrecMax	614	86	80	83	92	123	206	135	118	166	95	73	83	(16)
(17)		PrecMin	216	4	1	4	1	0	4	0	0	1	3	0	1	(17)
(18)		PrecStd	93	23	17	18	21	26	35	39	31	29	20	18	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	11.6	16.2	22.6	27.2	30.9	34.1	28.9	23.3	16.9	11.3	(19)	
(20)		MCWB	7.1	9.2	11.7	15.9	18.5	21.1	23.7	23.4	21.1	17.9	13.7	10.0	(20)	
(21)		2%	DB	6.7	8.8	12.0	19.3	25.0	29.0	31.6	31.3	25.6	21.2	14.5	9.2	(21)
(22)		MCWB	5.6	7.1	8.4	14.1	18.2	20.7	22.8	22.3	19.3	17.5	12.3	8.0	(22)	
(23)		5%	DB	5.3	6.7	10.2	17.1	23.6	27.6	30.2	29.8	24.2	19.6	12.7	7.8	(23)
(24)		MCWB	4.5	5.5	7.6	12.9	17.9	20.4	22.7	21.7	19.0	16.8	10.9	6.8	(24)	
(25)		10%	DB	4.1	5.3	8.7	15.3	22.3	26.4	29.2	28.6	23.0	17.6	11.1	6.1	(25)
(26)		MCWB	3.5	4.3	6.8	11.9	17.3	19.9	22.3	21.2	18.5	14.8	10.0	5.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.5	11.5	16.3	20.1	23.7	26.0	25.4	22.2	19.5	14.1	10.2	(27)
(28)		MDCB	8.3	11.4	15.7	21.7	24.1	27.8	29.3	30.8	25.9	21.9	15.9	11.0	(28)	
(29)		2%	WB	5.9	7.3	9.4	14.5	19.5	22.6	24.8	24.0	21.0	18.3	12.5	8.1	(29)
(30)		MDCB	6.6	8.6	11.2	18.2	23.4	27.3	29.4	28.9	23.9	20.5	14.3	9.1	(30)	
(31)		5%	WB	4.7	5.7	8.0	13.4	18.7	21.6	24.1	23.0	20.1	17.0	11.2	6.8	(31)
(32)		MDCB	5.2	6.6	9.5	16.4	22.6	26.1	29.3	27.9	23.1	19.3	12.3	7.7	(32)	
(33)		10%	WB	3.5	4.5	7.0	12.4	17.8	20.7	23.2	22.1	19.3	15.1	10.1	5.4	(33)
(34)		MDCB	4.0	5.2	8.5	14.9	21.7	25.1	28.6	27.3	22.2	17.3	11.0	6.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	4.1	5.0	5.0	6.0	7.4	8.1	8.5	8.7	7.3	6.5	4.5	4.4	(35)	
(36)		MCDBR	5.1	6.2	7.8	9.0	8.8	10.0	9.6	10.0	8.6	8.1	6.4	5.2	(36)	
(37)		MCWBR	4.4	4.9	5.1	5.2	4.7	5.4	5.2	5.5	4.9	5.3	4.5	4.5	(37)	
(38)		MCDBR	4.8	6.1	6.7	7.7	7.3	8.4	8.2	8.7	7.1	7.0	5.6	4.9	(38)	
(39)	MCWBR	4.3	5.0	4.9	4.7	4.4	5.4	5.2	5.9	4.5	5.1	4.3	4.4	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.376	0.403	0.446	0.416	0.421	0.443	0.444	0.416	0.383	0.360	0.346	(40)	
(41)		taud	2.294	2.228	2.201	2.114	2.297	2.318	2.266	2.248	2.290	2.347	2.343	2.307	(41)	
(42)		Ebn at Noon	727	777	814	811	853	848	823	805	793	764	707	680	(42)	
(43)		Edh at Noon	81	104	124	147	127	125	130	128	112	92	77	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.07	1.91	3.12	4.82	6.15	6.40	6.47	5.80	4.19	2.55	1.30	0.90	(44)	
(45)		RadStd	0.14	0.26	0.33	0.43	0.48	0.46	0.35	0.32	0.32	0.29	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.71	N/A	N/A	+0.98	N/A	N/A	-112	-158	+130	+83

Nomenclature: See separate page